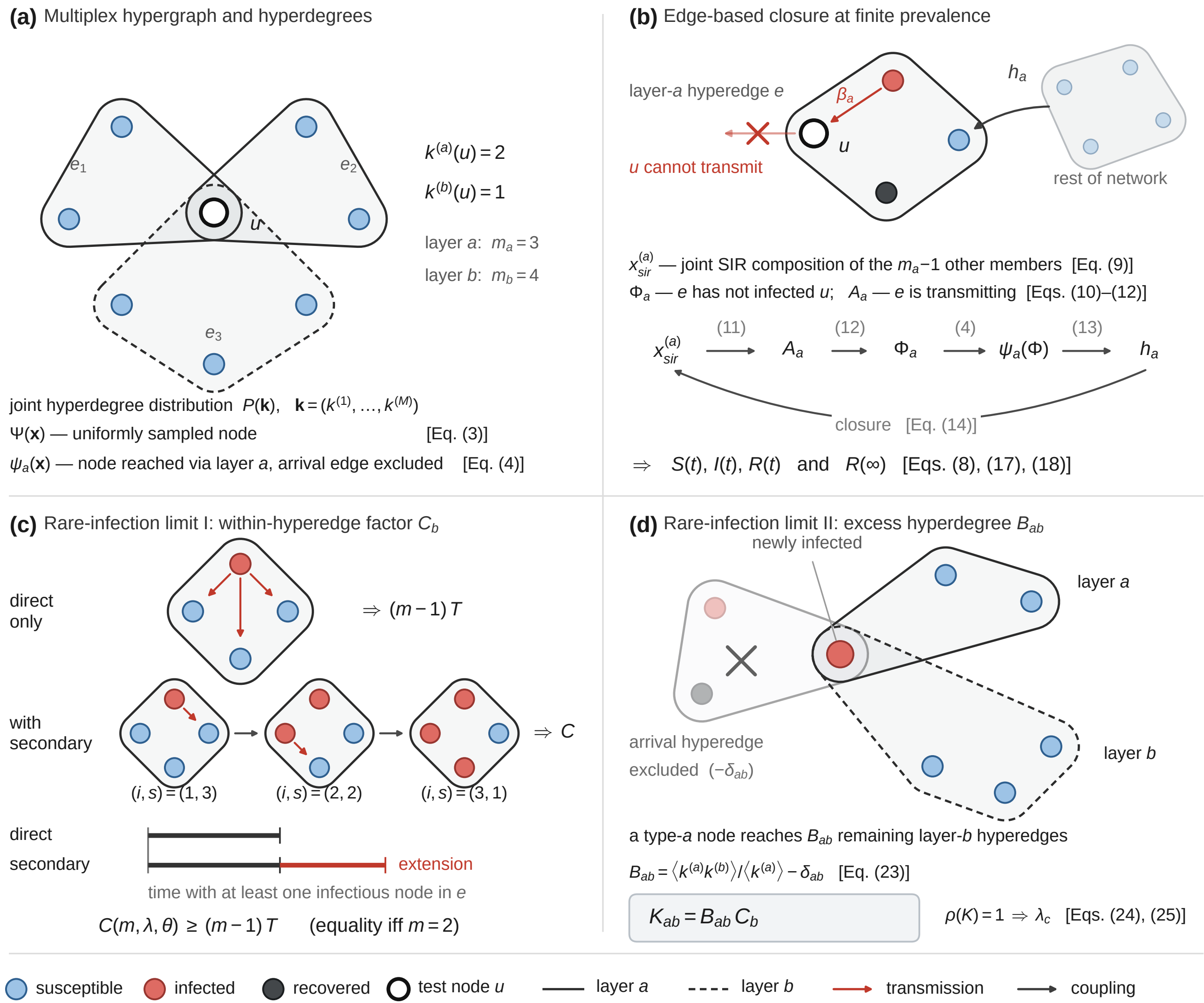

(a) Multiplex hypergraph and hyperdegrees
e_1
e_2
e_3
u
$k^{(a)}(u) = 2$
$k^{(b)}(u) = 1$
layer a: $m_a = 3$
layer b: $m_b = 4$
joint hyperdegree distribution $P(\mathbf{k})$, $\mathbf{k} = (k^{(1)}, \ldots, k^{(M)})$
$\Psi(\mathbf{x})$ — uniformly sampled node [Eq. (3)]
$\psi_a(\mathbf{x})$ — node reached via layer a, arrival edge excluded [Eq. (4)]
(b) Edge-based closure at finite prevalence
layer-a hyperedge e
β_a
u
u cannot transmit
h_a
rest of network
$x^{(a)}_{sir}$ — joint SIR composition of the m_a-1 other members [Eq. (9)]
Φ_a — e has not infected u; A_a — e is transmitting [Eqs. (10)–(12)]
$x^{(a)}_{sir} \xrightarrow{(11)} A_a \xrightarrow{(12)} \Phi_a \xrightarrow{(4)} \psi_a(\Phi) \xrightarrow{(13)} h_a$
closure [Eq. (14)]
$\Rightarrow$ $S(t)$, $I(t)$, $R(t)$ and $R(\infty)$ [Eqs. (8), (17), (18)]
(c) Rare-infection limit I: within-hyperedge factor C_b
direct only
$\Rightarrow (m-1)\,T$
with secondary
$\Rightarrow C$
$(i, s) = (1, 3)$
$(i, s) = (2, 2)$
$(i, s) = (3, 1)$
direct
secondary
extension
time with at least one infectious node in e
$C(m, \lambda, \theta) \geq (m-1)\,T$ (equality iff $m = 2$)
(d) Rare-infection limit II: excess hyperdegree B_{ab}
newly infected
layer a
layer b
arrival hyperedge excluded $(-\delta_{ab})$
a type-a node reaches B_{ab} remaining layer-b hyperedges
$B_{ab} = \langle k^{(a)}k^{(b)}\rangle / \langle k^{(a)}\rangle - \delta_{ab}$ [Eq. (23)]
$K_{ab} = B_{ab}\,C_b$
$\rho(K) = 1 \Rightarrow \lambda_c$ [Eqs. (24), (25)]
susceptible
infected
recovered
test node u
layer a
layer b
transmission
coupling